

IKERMIUARSUK, Greenland

WMO: 043820

Lat: 61.936N

Lon: 42.068W

Elev: 128

StdP: 14.63

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.2	13.6	-3.9	4.5	14.5	-1.0	5.2	16.7	62.7	21.8	58.9	23.0	19.3	300	1.029

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	51.1	43.5	49.0	42.3	47.2	41.4	44.9	48.7	43.8	47.1	42.8	45.7	8.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.9	40.9	44.9	41.8	39.2	44.1	40.7	37.5	43.2	17.7	49.0	17.1	47.3	16.6	45.8	52.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 52.3	45.5	40.0	DB	8.2	59.0	3.8	3.9	5.5	61.8	3.3	64.1	1.2	66.3	-1.5	69.2	(4)
(5)			WB	7.2	49.0	4.1	1.9	4.2	50.3	1.8	51.4	-0.5	52.5	-3.5	53.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	25.0	24.0	24.4	29.3	33.4	37.3	41.1	42.1	39.2	34.2	30.3	27.1		
	DBStd	7.49	5.10	5.37	5.12	4.07	2.65	3.62	3.92	3.26	3.11	3.62	3.43	4.48		
	HDD50	6454	776	729	793	620	515	382	278	245	323	491	591	711		
	HDD65	11928	1241	1149	1258	1070	980	832	742	710	773	956	1041	1176		
	CDD50	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	15.5	19.5	19.7	16.1	15.8	13.8	12.4	11.8	10.9	14.7	15.7	17.5	18.6		
	PrecAvg	46.4	5.2	4.4	3.4	3.1	2.6	3.0	3.0	3.6	4.4	4.0	4.8	4.8		
	PrecMax	61.8	11.2	10.4	6.3	6.3	6.5	4.7	4.8	5.9	7.8	8.7	8.5	14.0		
	PrecMin	37.0	1.4	1.4	0.6	1.1	0.8	0.9	0.9	1.0	2.4	1.3	2.1	0.8		
	PrecStd	6.4	2.2	1.9	1.3	1.2	1.1	1.0	1.0	1.2	1.3	1.7	1.7	2.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	35.9	37.5	41.6	44.8	51.8	55.2	54.4	50.2	46.5	41.1	38.7		
		MCWB	32.2	33.6	33.0	34.4	36.5	44.2	47.0	45.1	39.9	37.2	33.5	34.7		
	2%	DB	33.6	34.1	34.3	37.9	40.5	47.9	51.2	50.8	47.6	42.2	37.5	36.0		
		MCWB	31.3	32.7	32.4	33.3	35.3	41.8	44.2	43.2	39.0	36.6	33.3	32.9		
	5%	DB	32.3	32.4	32.8	35.7	38.7	45.1	48.9	48.7	45.4	40.5	35.9	34.3		
		MCWB	30.7	31.1	31.3	32.8	34.7	40.0	43.1	42.4	38.8	36.5	32.4	31.7		
	10%	DB	31.3	31.0	31.6	34.2	37.3	42.8	46.9	47.0	43.7	39.0	34.5	32.8		
		MCWB	29.7	29.6	30.2	32.1	34.0	39.2	42.0	41.8	38.8	35.2	31.6	30.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	34.9	35.0	36.4	38.8	45.1	47.8	46.5	44.3	40.5	36.8	36.5		
		MCDB	34.1	35.3	35.7	39.4	41.3	49.3	54.0	51.2	45.2	42.1	38.2	37.7		
	2%	WB	32.2	33.2	33.1	34.7	36.8	43.0	45.2	44.8	42.1	39.0	34.7	34.0		
		MCDB	32.8	33.6	33.7	36.2	39.0	46.5	49.5	48.4	44.1	40.7	35.8	35.3		
	5%	WB	31.3	31.5	32.0	33.7	35.6	41.1	43.9	43.7	41.0	37.6	33.5	32.5		
		MCDB	32.0	32.0	32.5	34.8	37.4	43.9	47.4	46.9	43.2	39.5	34.7	33.5		
	10%	WB	30.2	30.0	30.6	32.7	34.7	39.7	42.9	42.8	40.0	36.1	32.4	31.2		
		MCDB	31.0	30.8	31.6	33.8	36.4	42.2	45.9	45.5	42.2	38.3	33.7	32.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.4	7.0	7.1	6.6	7.3	8.0	7.4	5.9	5.4	4.8	4.8		
		MCDBR	6.3	7.0	7.6	9.6	9.5	11.3	11.7	10.4	8.7	7.5	6.8	6.7		
		MCWBR	5.7	6.2	6.5	6.4	6.0	6.5	6.6	5.6	4.8	4.6	4.6	4.9		
	5% WB	MCDBR	6.1	6.1	6.3	7.0	7.9	9.6	10.6	8.9	6.5	6.6	5.8	5.6		
		MCWBR	5.9	6.1	5.9	5.1	5.4	6.4	6.7	5.6	4.6	4.6	4.7	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.277	0.284	0.323	0.333	0.341	0.336	0.326	0.307	0.277	0.224	0.373		
	taud		2.014	2.201	2.295	2.190	2.211	2.290	2.377	2.430	2.418	2.332	2.084	2.769		
	Ebn at Noon		134	218	265	273	280	279	279	272	256	219	146	99		
	Edh at Noon		12	21	27	36	39	37	33	29	24	18	10	7		
(44) All-Sky Solar Radiation	RadAvg		81	305	805	1439	1938	2005	1892	1459	905	445	132	43		
	RadStd		8	33	50	115	138	177	98	99	58	43	11	3		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	+1.57	+5.22	N/A	+1.10	N/A	-1.13	-490	-464	N/A	N/A

Nomenclature: See separate page